Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cylinders and spheres, and volume. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	5
Topic	Three dimensional shapes II
Subtopic	Three dimensional figures: properties of cylinders and spheres, and volume
Theme	—
Curriculum objectives	• identify properties of cylinders and spheres • find the volume of a cube, a cuboid and a prism
Previous knowledge	• properties of cubes and cuboid • properties of pyramids and cones
Lesson sequence	• identify properties of cylinders and spheres — the teacher displays shapes of cylinders and spheres; the student identify the properties; the teacher draws the shapes on the board and adds to the properties already mentioned by the students. Set for the students: 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces. • find the volume of a cube, a cuboid and a prism — the teacher explains how to derive the formula for volume of cube and cuboid, and works the volume of a prism as area of base times height; the student listen and ask questions where they do not understand; the student copy worked examples into their notes
Instructional materials	• shapes of cylinders and spheres.
Evaluation	• Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high. • Calculate the volume of cuboid with dimensions 5m by 2m by 3m. • What is the volume of a cube of 4cm × 4cm × 4cm. • A rectangular tank hold 6m long by 2m wide holds $36m^3$ of water. How deep is the water in the tank.

Lesson objectives

1.	identify properties of cylinders and spheres
2.	find the volume of a cube, a cuboid and a prism

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Three dimensional figures: properties of cylinders and spheres, and volume

Class: JSS1 **Subject:** Mathematics **Week:** 5

Section 1: Cylinders, Spheres, and Finding Volume

Two solid shapes you meet all the time are the cylinder and the sphere. Let's look closely at what makes each one special, so you can pick them out anywhere.

A cylinder has two identical flat circular faces, one at the top and one at the bottom. These two circles are called the bases. The two bases are joined by a single curved surface that wraps all the way around. A cylinder has no sharp points (no vertices), but it does have two edges — the two circular rims where the curved surface meets each base. Because of that curved surface, a cylinder can roll along the ground, but only in a straight line (it rolls end over end, not in all directions). Think of a tin of baked beans or a drinking glass.

A sphere is perfectly round — every point on its surface is exactly the same distance from its centre. That fixed distance is called the radius. A sphere has no flat faces, no edges, and no vertices at all. Because it has only one continuous curved surface, a sphere rolls in every direction. Think of a football or a tennis ball.

Here is a quick way to tell them apart: a sphere is the only shape that rolls freely in every direction and has no flat face, no edge, and no corner. A cylinder has exactly two flat circular faces, two circular edges, and no corners.

Volume is the amount of space a solid takes up, and we measure it in cubic units such as cm³ or m³. Three box-like solids have a simple volume formula you can rely on.

A cube is a solid where every edge is the same length. If its edge is s, then its volume is:
Volume of cube = s × s × s = s³

A cuboid (a rectangular box) has three different edges — length l, width w and height h. Its volume is:
Volume of cuboid = l × w × h

A prism is any solid that has the same shape all the way through from one end to the other. The two ends are identical polygons, and they are joined by flat rectangular faces. The volume of any prism is:
Volume of prism = area of the cross-section × length

The cross-section is the shape of the two identical ends — for example, if a prism has a triangular end, you first find the area of that triangle and then multiply by the prism's length. So a cuboid is actually a prism whose cross-section is a rectangle, and the rule "area of cross-section × length" gives the same result as l × w × h.

Worked Example 1 — Properties of a cylinder
A tin of soup has two flat circular ends and one curved surface that joins them.
How many faces, edges and vertices does it have?
The two circular ends are flat faces, and the curved surface is not flat but it is still one surface. So there are 3 faces in total. There are two circular edges — the rim where the curved surface meets each circular base. There are no vertices, because no flat faces meet at a point.

Worked Example 2 — Volume of a cube
Find the volume of a cube with edge 4 cm.
Volume = s × s × s = 4 × 4 × 4 = 64 cm³.

Worked Example 3 — Volume of a cuboid
A box is 5 cm long, 3 cm wide and 2 cm high. Find its volume.
Volume = l × w × h = 5 × 3 × 2 = 30 cm³.

Worked Example 4 — Volume of a prism
A triangular prism has a triangular end whose area is 8 cm² and a length of 6 cm. Find its volume.
Volume = area of cross-section × length = 8 × 6 = 48 cm³.

Practice

1. A ball has only one continuous curved surface and no flat faces, no edges and no vertices. Is the ball a cylinder or a sphere? Give a reason.
2. A drinking glass is shaped like a cylinder. State the number of flat faces, the number of edges and the number of vertices it has.
3. Find the volume of a cube whose edge is 5 cm.
4. A cuboid is 8 cm long, 4 cm wide and 3 cm high. Find its volume.
5. A prism has a triangular cross-section with area 12 cm² and a length of 7 cm. Find its volume.

Section 2: Volume of Cubes, Cuboids and Prisms

Volume is the amount of space a solid takes up. It is measured in cubic units, like cubic centimetres (cm³) or cubic metres (m³).

A **cube** has every edge the same length. If the edge is a, then:

$$\text{Volume} = a \times a \times a = a^3$$

A **cuboid** (a rectangular box) has three different edge lengths: length l, width w and height h. Its volume is:

$$\text{Volume} = l \times w \times h$$

A **prism** is a solid with the same shape (a face) all the way through. A cube and a cuboid are both prisms. For any prism, the volume is found by multiplying the area of the end face (the cross-section) by the length (or height):

$$\text{Volume} = \text{area of cross-section} \times \text{length}$$

So the same rule works for a cube, a cuboid and any other prism: find the area of the end face, then multiply by how long the prism is.

Example 1: a cube

A cube has edges of length 5 cm. Find its volume.

[\text{Volume} = a^3 = 5 \times 5 \times 5 = 125]

The volume is **125 cm³**.

Example 2: a cuboid

A cuboid is 8 cm long, 3 cm wide and 4 cm high. Find its volume.

[\text{Volume} = l \times w \times h = 8 \times 3 \times 4 = 96]

The volume is **96 cm³**.

Example 3: a prism

A triangular prism has a triangular end face with area 12 cm² and is 7 cm long. Find its volume.

[\text{Volume} = \text{area of cross-section} \times \text{length} = 12 \times 7 = 84]

The volume is **84 cm³**.

Practice

1. A cube has edges of length 6 cm. What is its volume?
2. A cuboid is 10 cm long, 4 cm wide and 3 cm high. What is its volume?
3. A prism has an end face with area 15 cm² and is 8 cm long. What is its volume?

Check your work

Attempt every question before you read this page.

Cylinders, Spheres, and Finding Volume

1. It is a sphere, because a sphere has no flat faces, no edges and no vertices, and it rolls freely in every direction.

   A cylinder has two flat circular faces and two circular edges, but a ball has none of these. A shape with no flat face, no edge and no corner is a sphere.

2. 2 flat faces, 2 edges, 0 vertices.

   A cylinder has two flat circular bases joined by one curved surface. That gives 2 flat faces, the two circular rims where the curved surface meets each base give 2 edges, and there are no corners where faces meet in a point, so 0 vertices.
3. 125 cm³.

   Volume of a cube = s × s × s = 5 × 5 × 5 = 125 cm³.
4. 96 cm³.

   Volume of a cuboid = l × w × h = 8 × 4 × 3 = 96 cm³.
5. 84 cm³.

   Volume of a prism = area of cross-section × length = 12 × 7 = 84 cm³.

Volume of Cubes, Cuboids and Prisms

1. 216 cm³

   Volume of a cube = edge × edge × edge = 6 × 6 × 6 = 216. The volume is 216 cm³.
2. 120 cm³

   Volume of a cuboid = length × width × height = 10 × 4 × 3 = 120. The volume is 120 cm³.
3. 120 cm³

   Volume of a prism = area of cross-section × length = 15 × 8 = 120. The volume is 120 cm³.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-b1e989fa	**Date**	
Subtopic	Three dimensional figures: properties of cylinders and spheres, and volume	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____